PRISMA 2020 Eligibility Stage: Quality Criteria Scoring — Section 6.0 (Empirical Evidence of Generalisation Failure under Distributional Change)

Report prepared 20 August 2026

1. Scope of This Scoring Pass

This report extends the Step 4 eligibility scoring to the full candidate set retrieved for Section 6.0 (Empirical Evidence of Generalisation Failure under Distributional Change) of the review, comprising all 13 full-text articles retained for this subsection following Step 3 screening. Each article was scored honestly against its full text rather than against title and abstract alone.

QC1 (problem statement and scope clarity) was scored against fit to this subsection's specific purpose, namely documenting empirical instances of cross-environment, cross-time, or cross-deployment generalisation failure in machine learning-based intrusion detection, as distinct from proposing a detection or adaptation mechanism, which is the subject of Sections III to V, or reviewing the field secondarily, which is a different research question again. An article can present a clear problem statement for a closely related but distinct question, such as remediating a generalisation gap via feature selection, or synthesising others' cross-dataset findings within a broader survey, and still score partially on QC1 here, because the object of appraisal in Section VI is empirical documentation of the failure itself.

QC4 (empirical extraction detail) was scored as 0 wherever an article's full text contained no drift-type, distributional-component, or adaptation-mechanism data extractable into this review's own extraction fields, regardless of how much extraction detail the article offered for its own, different research question. This reflects a consistent, review-specific application of the criterion rather than a per-article adjustment.

QC2 (methodological transparency) and QC3 (venue indexing and citability) were scored on their ordinary, article-intrinsic basis, since these criteria measure the rigor and citability of the work itself rather than its fit to this particular review.

2. Quality Assessment Criteria Applied

Table 1 reports the four quality criteria (QC1 to QC4) applied in this eligibility round, together with the scoring scheme used for each. A quality score is computed as the sum of applicable criterion scores divided by the number of applicable criteria, scaled to a 0 to 4 range.

Table 1. Quality assessment criteria applied to Section 6.0 candidate articles

Code	Criterion	Scoring
QC1	Problem statement and scope clarity	Yes (1) / Partial (0.5) / No (0)
QC2	Methodological transparency	Yes (1) / Partial (0.5) / No (0)
QC3	Venue indexing and citability	Yes (1) / Partial (0.5) / No (0)
QC4	Empirical extraction detail (where applicable)	Yes (1) / Partial (0.5) / No (0) / N/A

Articles published in 2020 or later were retained where their total quality score met or exceeded 2.5 out of 4. Articles published before 2020 were retained where their total quality score met or exceeded 1.5 out of 4.

3. Results

Of the 13 full-text articles assessed for Section 6.0 against the quality criteria in Section 2, 8 articles met or exceeded their applicable threshold and were retained, while 5 articles fell below their applicable threshold and were excluded. Table 2 lists each assessed article together with its per-criterion and total quality scores, applicable threshold, and eligibility decision, sorted with included articles listed first, followed by excluded articles.

Table 2. Quality criteria scoring results for Section 6.0 candidate articles, sorted by decision

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
1	Explainable cross-domain evaluation of ML-based network intrusion detection systems (2023)	2023	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
2	Inter-dataset generalization strength of supervised machine learning methods for intrusion detection (2020)	2020	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
3	Benchmarking and Cross-Dataset Evaluation of AI-Based Intrusion Detection Systems for Smart City IoT Networks (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
4	Evaluating the Generalization Gaps of Intrusion Detection Systems Across DoS Attack Variants (2025)	2025	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
5	Concept Drift Analysis by Dynamic Residual Projection for Effectively Detecting Botnet Cyber-Attacks in IoT Scenarios (2022)	2022	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**

No.	Citation	Year	QC1	QC2	QC3	QC4	Total	Threshold	Decision
6	FORT-IDS: A federated, optimized, robust and trustworthy intrusion detection system for IIoT security (2025)	2025	1.0	0.5	1.0	1.0	**3.5**	≥2.5	**Include**
7	Dataset-centric evaluation of federated intrusion detection models in IoT networks (2026)	2026	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
8	DI-NIDS: Domain invariant network intrusion detection system (2023)	2023	1.0	1.0	1.0	1.0	**4.0**	≥2.5	**Include**
9	SHAP-Guided Feature Selection for Cross-Dataset Generalization in Network Intrusion Detection Systems (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
10	Machine Learning in Network Intrusion Detection: A Cross-Dataset Generalization Study (2024)	2024	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
11	Toward Generalization and Interpretable Deep Learning-Based Intrusion Detection System for Heterogeneous Network Environments (2026)	2026	0.5	0.5	1.0	0.0	**2.0**	≥2.5	**Exclude**
12	A Decentralized Federated WGAN-GP Framework for Anomaly Detection in Industrial Control Systems (2026)	2026	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**
13	A Systematic Review of Intrusion Detection Systems for Internet of Medical Things: Performance, Efficiency, Explainability, and Generalization (2026)	2026	0.0	1.0	1.0	0.0	**2.0**	≥2.5	**Exclude**

This retained set of 8 articles constitutes the eligible pool for Section 6.0 (Empirical Evidence of Generalisation Failure under Distributional Change) and forms part of the input to the synthesis and thematic analysis stage.

4. Data Integrity Notes

Each article was scored against the four quality criteria on the basis of its full text, applying the review-specific interpretation of QC1 and QC4 set out in Section 1 above. The retain or exclude decision was applied strictly according to the threshold in effect for that article's publication year, namely 2.5 out of 4 for articles published 2020 or later, or 1.5 out of 4 for articles published before 2020. No article was retained or excluded on any basis other than its quality score relative to its applicable threshold.

5. Document Preparation

This report has been prepared in accordance with the quality assurance procedure prior to proceeding to the synthesis stage.

Prepared by
Name: RAZIMAN ZAKARIA